Technical & Functional Specification (TSD/FSD)

Tech Stack

Layer	Tool	Notes
Database	Supabase (Postgres 15)	Row-Level Security, Auth, auto-generated Data API
LLM	Google Gemini API	Free-tier; Anthropic Claude API available as a paid alternative
Backend scripting	Python 3.11+	`supabase-py`, `python-dotenv`, `pandas`, `openpyxl`, `google-generativeai`
Frontend	Streamlit	Hosted on Streamlit Community Cloud
Scheduling	GitHub Actions	Cron-based, for future automated API pulls
Version control	GitHub	`arunmanimal/iThink_v2`

Data Model (as built)

Table	Purpose
`tenants`	One row per customer organization
`tenant_users`	Maps auth users to tenants with a role (admin/analyst/viewer)
`brands`	Own brand + competitor brands per tenant
`brand_aliases`	Alternate names per brand, fed to the LLM prompt for matching
`data_sources`	Registered ingestion sources per tenant (excel/sprinklr/brandwatch/data_lake)
`business_taxonomy`	8 top-level domains (Sales, Service, Product, QAD, NPD, Connected Vehicle, Corporate, Dealer Experience)

Table	Purpose
`attribute_taxonomy`	168 consumer attributes under the 8 domains, tagged ICE / EV / ICE & EV
`signal_taxonomy`	80 AI signal tags (Sentiment, Emotion, Intent, Trend, Impact, Severity, Importance, Customer Journey, Competitive Context, Competitor & Market Intelligence)
`conversations`	Ingested raw conversations + `raw_metadata` jsonb
`conversation_attribute_tags`	Many-to-many: conversation ↔ attribute
`conversation_signal_tags`	Many-to-many: conversation ↔ signal
`conversation_brands`	Many-to-many: conversation ↔ brand

Functional Modules

Module	Script	Status
Taxonomy seeding	`scripts/seed_taxonomy.py`	Done
Brand & alias setup	SQL inserts	Done
Ingestion adapter	`scripts/ingest_excel.py`	Planned — reads Sprinklr export, maps to `conversations` + `raw_metadata`, dedupes on `source_message_id`
Extraction pipeline	`scripts/extract_tags.py`	Planned — batches conversations, calls Gemini with taxonomy + brand-alias context, writes attribute/signal/brand tags
Aggregation job	TBD	Planned — rolls tagged conversations into brand × attribute × period scores
Dashboard	`app/dashboard.py`	Planned — Streamlit, pending attributes-for-display, layout, and color scheme from stakeholder
Auth & onboarding	TBD	Planned — Supabase Auth wiring, tenant sign-up flow, RBAC

Non-Functional Specs

- Idempotency via unique constraints (`source_message_id` per tenant; taxonomy name uniqueness)
- RLS enforced on every tenant-scoped table; `service_role` key used only in trusted backend scripts, never client-side
- Free-tier constraints observed: Supabase free tier (500MB DB, 50k MAU), Gemini free-tier rate limits
- All secrets (`SUPABASE_SERVICE_ROLE_KEY` , `GEMINI_API_KEY`) kept in `.env` , excluded from git via `.gitignore`